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Appendix

We open source our code at `https://github.com/GeMoLLM/GeMO`.

A Sampling Parameters

1. Temperature: Concretely, the temperature parameter T determines the "dispersion" of the probability distribution over the next token. Mathematically, the probability of a token w being generated is given by the softmax function: $P(w|x) = \frac{\exp(s(w|x)/T)}{\sum_{w'} \exp(s(w'|x)/T)}$ where $s(w|x)$ is the unnormalized log-probability of the token w given the context x. Increasing the temperature leads to a more flat probability distribution and increases the likelihood of sampling from less probable tokens, resulting in more diverse generations. This can also be understood from the perspective of entropy of the next token, $H(W) = -\sum_w P(w|x) \log P(w|x)$, where it can be seen that the increase of entropy directly follows.

2. Top-p: The top-p parameter [35] ($p \in (0, 1]$) controls the randomness of the generations by limiting the range of tokens considered. Specifically, it considers the smallest subset (consisting of the top probability tokens) whose cumulative probability exceeds the threshold p. A smaller p encourages the model to sample from a more focused set of likely tokens, while a larger p allows sampling from a broader range and thus increases randomness ($p = 1$ basically means no restriction on the vocabulary).

Different platforms have different default values for T and top-p. `GPT-3.5-turbo-instruct (0914)` web version adopts $T = 0.8$ [49]. The default values in OpenAI APIs are $T = 1.0$ and $p = 1.0$ [50].

3. Generation Length: This parameter (`max_new_tokens`) dictates at most how many tokens the model should generate before it stops. We used 500 for book review generation and 2048 for code generation.

B Additional Details: Book Reviews

B.1 Construction of the Source Dataset

To ensure we picked popular books that the LLMs know about, we filtered the books according to the number of reviews they have. Constraining the number of reviews to be between 1,000 and 2,500, we obtained 750 books. We further filtered out books with non-English titles; we conducted this filtering because some downstream LLMs (e.g., sentiment classifiers) are not multilingual, and as a result can not analyze the generated non-English reviews. To ensure high quality of the reviews, we further filtered the review by length, constraining the length of each review to be between 300 and 700 words. After this step of filtering, we sampled 10 reviews per book. Eventually, we obtained a dataset of 742 books, where each book comes with 10 reviews.

B.2 Names of the Celebrities Used in Prompt 2

We prompted `GPT-4` to provide a list of celebrities suitable for writing diverse book reviews. The list of names are as follows: Trevor Noah, Janelle Monáe, Yuval Noah Harari, Serena Williams, Reshma Saujani, Neil deGrasse Tyson, Margaret Atwood, David Attenborough, Malala Yousafzai, Jordan Peele.

B.3 Text Processing for Analyzing Wording Choice

We first concatenated all documents and converted them to lowercase to standardize the text. We then expanded contractions (e.g., "don't" to "do not"). We then tokenize the text with `word_tokenize()` from the NLTK library [51], and removed the punctuation. We continued by filtering out non-alphabetic tokens to focus solely on words. Lastly, we lemmatized tokens via `WordNetLemmatizer()` to their base forms, aiding in consistent frequency analysis. These steps are essential for minimizing textual noise and ensuring the reliability of our word frequency assessments.

C Additional Results: Book Reviews

C.1 Filtered-out Reviews

We established a perplexity threshold of 20 to filter out low-quality reviews. To validate our choice, we randomly sampled reviews with perplexity scores at different intervals and manually inspected them. For clarity, instead of presenting the entire review text, we selectively extracted chunks that exemplified the low-quality nature of the sampled reviews.

Perplexity in $(20, 25]$: This page-turning, and fast-paced gripping story makes excellent use both tropes and novel ideas by employing both like, for instance, a dystopian setting which uses an exaggeratory ruminating setting so typically found in series belonging to this genre, only that here the author‘s unique skill at weaving such plot-lines together in a novel that feels fresh more like Harry Potter but with an 800word length page each chapter instead of JUST 200word paragraph at the end of an excrutiatingly slow chapter.

Perplexity in $(25, 30]$: That and artist Adrian Alphona bringing the world-sprawling amazing action and gorgeous characters from beautifully rendered backgrounds in color by vivid and dazzling color is what makes the graphic novel in itself even more delightful. With each color panel bringing an explosion of color onto every page no panel is left the same no character or page lacking the same amount of energy vivid and gripping from the moment I turned on. the illustrations are bright beautiful detailed it truly does it live upto its tag line and then some a hero like no other a book like no other definitely worth the dive.

Perplexity in $(30, 35]$: One would wonder if she lived some of this amazing journey to unhidden holes in the ground or lived the stories and characters by the holes they created one will see how amazingly her fantastic works has a way t connect the reader to a place which feels so real and fascinating! Holes by Loues Erdrich has a beautifully constructed a magical and a world which transports and takes readers like on a journey thrifting us so many magical and real places which we might not have an opportunity if it wasn't for the eyes and minds eye of Louise Erdrech .

Perplexity in $(35, 40]$: It made me reflect for a deeper appreciation on both the power of memory in creating some of literature and art's most significant works or cultural touchstone masterpieces such as those listed that our lead female artist was known to adore and how that and the themes that author – a most wonderful story teller by the way, by way of characters that i wanted or did want to spend more time with and learn the most in depths exploration of, and not all can reach it. And so a deeper appreciation for the art of storytelling from great narratives by great narrators? The Little Paris Bookp shop does achieve an astounding success in that regard, I am thrilled I'd like to shout it out to any and All in ear shot about it!

C.2 Generation Results at Higher Randomness

We experimented with even higher randomness at $T = 1.5$ motivated by the observation that increasing the randomness does help to increase the diversity. However, as we show in Fig. 6, even at high randomness, there is still a huge gap between the source and the generations.

Moreover, we notice a significant degradation of the generation quality as a result of the increased randomness. We present in Table 1 the average number of valid generations for two models under various sampling parameters. The table shows that the valid number of generations rapidly drops as the randomness increases, particularly at $T = 1.5$; the implication is that such a high randomness setting basically cannot be adopted for practical use.

C.3 Topic Shifts

We present in Fig. 7 the results of the unconditional topic distribution. Across all three settings for comparison, we observe highly similar distribution shift signified by the over- and under-representation of certain topic groups. Concretely, the topic group 15 with the keywords "grace, rob, nick, anna, house, discovers, realizes, killed, confronts, killer" are under-represented, potentially because certain words like "rob" and "kill" are *eliminated* from the output as a result of RLHF. On the other hand, the topic group 333 with the keywords "handmaid, novels, novel, writers, tale, book, books, literary, novellas, synopsis" and the topic group 666 with the keywords "nonfiction, bestseller, novelist, autobiography, novels, memoir, author, memoirs, paperback, novel" are over-represented. The over-representation of topic group 333 in `Vicuna-13b` is much more severe than `Llama-2-chat`, as can be seen from the 3rd subfigure; this could be because a higher exposure to relevant materials during its fine-tuning.

C.4 Pre-trained Model

The aligned model is obtained from performing supervised fine-tuning followed by RLHF alignment tuning on the basis of the pre-trained model [32]. We compare the performance of the aligned model llama-2-13b-chat with the pre-trained model llama-2-13b and present the results in Fig. 8.

From the 1st and 2nd subfigure, we observe that the pre-trained model is much more diverse than the aligned model; the pre-trained model is even very close to the source data (see the dark purple result tagged as $T = 1.2, p = 1.0$ (c)).

The 3rd subfigure delivers two messages. First of all, there still exists divergence from the pre-trained model and the source in terms of the covered topics. We attribute this divergence we observe between the pre-trained model and the source data to the difference between the source data we use (the Goodreads dataset) and the ground-truth training data (a much broader corpus which we have no information about). We regard the Goodreads dataset as a proxy of the ground-truth, but intrinsically Goodreads is smaller and does not fully accurately represent the groundtruth training distribution, especially in nuanced attributes like the unconditional topic distribution measured on all samples. Second, the direction and trend of changes in the aligned model is not completely the same as the aligned models, e.g., the topic group 15 containing "rob" and "killer" is not inhibited as much as in the aligned models, potentially due to RLHF.

C.5 Results of OpenAI models

We repeat the same set of experiments on the proprietary OpenAI models GPT-3.5-turbo-instruct (0914) and GPT-4-turbo (0125). We present the results in Fig. 9.